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DETAIL SPECIFICATION

3000 GALLON AERIAL BULK FUEL DELIVERY SYSTEM BLADDER

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for a 3000 Gallon Aerial Bulk Fuel Delivery System (ABFDS) Bladder, used with the C-5, C-17, and C-130 aircraft.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to:
U.S. AIR FORCE, AFLCMC/WN-SEV, 380 Richard Ray Blvd, Robins AFB GA 31098 or emailed to SPEC99@us.af.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

AMSC N/A FSC 5430

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FEDERAL STANDARDS

FED-STD-123	Marking for Shipment

COMMERCIAL ITEM DESCRIPTIONS

A-A-59326/4	Coupling Half, Male by Flange, Type IV
A-A-59326/8	Coupling Half, Female by Flange, Type VIII
A-A-59326/10	Coupling Half, Cap, Dust, Type IX
A-A-59326/11	Coupling Half, Plug, Dust, Type X
A-A-59326/12	Reducer, Male by Female and Female by Male, Type XI

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-83133	Turbine Fuels, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-T-52983	Tanks, Fabric, Collapsible: 3,000,10,000,20,000, and 50,000 Gallon, Fuel

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	Military Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	System Safety
MIL-STD-1472	Human Engineering
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE DEPARTMENTAL PUBLICATION

AFI 32-7086	Hazardous Materials Management

(Copies of these documents are available online at http://www.e-publishing.af.mil.)

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ARMY DRAWING

ARMY 11-1-2780	Platform, Aerial Delivery Type V

(Copies of these documents are available from U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11Mile Road, Warren, MI 48397-5000 or can be requested by sending an email to: usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil/).

DEPARTMENT OF DEFENSE MANUALS AND PUBLICATIONS

DoDM 4140.27	DoD Shelf-Life Management Manual

(The DoD Manual is available online at http://www.esd.whs.mil/ .)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI H35.1	Alloy and Temper Designation Systems for Aluminum

(Copies are available from the American National Standards Institute, 1899 L Street, NW, 11th Floor, Washington DC 20036 - Electronic copies may be obtained at http://www.ansi.org.)

ASTM INTERNATIONAL (ASTM)

ASTM A276	Standard Specification for Stainless Steel Bars and Shapes
ASTM-B85/B85M	Standard Specification for Aluminum Alloy Die Castings
ASTM B221	Standard Specification for Aluminum and Aluminum Alloy Extended Bars, Rods, Wire, Shapes and Tubes
ASTM D975	Oils, Diesel Fuels
ASTM D1655	Standard Specification for Aviation Turbine Fuels

(Copies of these documents are available from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), a Bladder (or Bladders) shall be subjected to first article inspection in accordance with 4.2.

3.1.1 Bladders for testing. Bladders selected as testing samples are in addition to the quantity required for delivery and shall remain the property of the contractor after completion of testing. All Bladders selected, as samples for destructive testing must be completed samples.

3.2 Materials. The materials and components shall conform to applicable specifications, standards, and patterns required herein. Unless otherwise specified, nonmetallic materials used in construction of the assembly shall not have been manufactured more than 24 months prior to the date of delivery, except for rubber products, which shall have a cure date not more than 12 months prior to the delivery date. However, when a definite material is not specified, a material shall be used which will enable the Bladders to meet the performance requirements of this specification.

3.2.1 Material deterioration, prevention, and control. The Bladder shall be fabricated from compatible materials, inherently resistant or treated to provide protection against the various forms of deterioration that may be encountered in the operational and storage environments to which it may be exposed.

3.2.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless appropriately protected against galvanic corrosion.

3.2.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with components and subcomponents, and shall make information available upon request to the Government.

3.2.1.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the materials meet or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Coated fabric. The coated fabric shall be free from blisters, holidays or pinholes, and shall show no signs of coating delamination. The coated fabric shall withstand the effects of all environmental elements stated herein without damage or deterioration of performance requirements.

3.2.2.1 Coating compounds. Compounds used for the formulation of Acrylonitrile-Butadiene coatings shall be suitable for use with hydrocarbon fuels containing up to 40 percent aromatics. The coating compounds shall be resistant to fungus, weathering, ozone, hydrolytic attack, and temperatures up to 160 °F.

3.2.2.2 Nylon cloth. The nylon cloth of the coated fabric shall be high tenacity and heat and light resistant. The nylon cloth shall be free from any imperfections affecting strength, coating adhesion, or coating thickness.

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3.2.2.3 Weathering resistance. When tested as specified in 4.5.7, the nylon cloth shall have a retained breaking strength that is not less than 45 percent of its initial breaking strength in warp and fill.

3.2.3 Fill/evacuate and vent nozzle material. Alloy and temper designations of wrought aluminum shall be in accordance with ANSI-H35.1. Cast aluminum alloy shall conform to ASTM-B85/B85M. Mold and sand castings are permissible.

3.2.4 Fill/evacuate and vent flange material. Alloy and temper designations of wrought aluminum shall be in accordance with ANSI-H35.1. Cast aluminum alloy shall conform to ASTM-B85/B85M. Mold and sand castings are permissible.

3.2.5 "D" rings material. The "D" ring material shall be equivalent to ASTM-A276 Type 304L stainless steel rod, low carbon alloy steel that allows welding.

3.3 Design and construction. The Bladder shall be constructed in accordance with the design in Appendix A. It is a built-up assembly of a number of constituent parts. These parts shall be joined by a leak-proof bonding process, which results in a homogeneous Bladder equipped with fourteen (14) diagonal forward restraint straps, ten (10) lateral double "D" tie down rings, a fore and aft tow ring, and eight (8) lift handles. The Bladder shall also equipped with a vent assembly and fill/evacuate assembly. Internal to the Bladder shall be three (3) rows of fluid flow restrictors (synonymous with surge suppresser) tied to the internal Bladder with nylon tie cord.

3.3.1 Bladder and associated component design.

3.3.1.1 Basic Bladder. The basic Bladder shall consist of up to six (6) sections orientated laterally or longitudinally, joined by a vulcanized bonding process. The composition of the basic Bladder material shall be at a minimum Acrylonitrile-Butadiene Rubber (NBR) and two (2) plys of nylon fabric inter-layered to form a layered sandwich construction. Additional components in the fabric build are permitted with the approval of the Government. The Bladder perimeter dimensions shall be 110 inches wide, 242 inches long, and approximately four (4) inches high when collapsed. The Bladder mid section dimensions shall be approximately 100 inches wide, 235 inches long, and 36 inches high when filled to normal capacity.

3.3.1.2 Double “D” tie down rings. There shall be ten (10) double "D" tie down rings located on the upper Bladder external surface. The rings shall be attached to the Bladder's longitudinal centerline by means of a coated NBR nylon strap, which is bonded to a patch, which, in turn, is bonded to the Bladder surface.

3.3.1.3 Lift handles. The Bladder shall be equipped with eight (8) lift handles located below the natural fold line of the Bladder. The straps shall be bonded to a patch, which is, in turn, bonded to the external Bladder surface.

3.3.1.4 Fill/evacuate assembly. The fill/evacuate assembly shall consist of two subassemblies, which are mated with mounting bolts utilizing an O-ring for the sealing medium:

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a. Subassembly No. 1. Subassembly 1 shall be the fill/evacuate nozzle which allows the attachment to the Bladder on one side and the cap sealing of the nozzle or hose attachment point on the other. The fill/evacuate nozzle shall contain a foot feature on the lower surface of the mounting plate. The foot feature shall consist of four(4) to six (6) pedestals with diagonal channels across the pedestals. This feature enables the fill/evacuate nozzle to come into contact with the lower internal surface of the Bladder during fuel evacuation without allowing the lower surface to be sucked up into the suction port.

b. Subassembly No. 2. Subassembly 2 shall be the fill/evacuate flange which is made integral with the Bladder's surface by the use of adhesive or vulcanized bonding process and NBR coating to form a leak-proof attachment.

3.3.1.5 Vent assembly. The vent assembly shall consist of two (2) subassemblies, which are mated with mounting bolts utilizing an O-ring for the sealing medium.

a. Subassembly No. 1. Subassembly 1 shall be the vent nozzle, which allows the attachment to the Bladder on one side and the cap sealing of the nozzle or the attachment of the anti-siphoning valve on the other.

b. Subassembly No. 2. Subassembly 2 shall be the vent flange, which is made integral with the Bladder's surface by the use of adhesive or vulcanized bonding process and NBR coating to form a leak- proof attachment.

3.3.1.6 Reed valve vapor eliminator. The Bladder shall be equipped with an automatically actuated reed valve vapor eliminator. The vapor eliminator shall attach to the vent assembly. It shall be capable of venting vapor through a ¾ inch ID hose that is attached to the aircraft overboard vent.

3.3.1.7 Fluid flow restrictors. The Bladder shall be equipped with three (3) internal rows of fluid flow restrictors, which control fuel surge as illustrated in Figure 2-A. These restrictors shall be trapezoidal sections of material, which are a sandwich design with a stiffener to allow them to stand away from the Bladder's internal skin as shown in Figure 3-A. There shall be twenty (20) Sections in each row arranged alternately to restrict flow in both longitudinal directions. Each section of material shall be tied to the Bladder's internal surface by means of an interlaced nylon cord utilizing eyelet-protected openings for the lacing. The eyelets shall be present on the restrictor sections and mating ribs of Bladder material, which enable a continuous ring of restrictors to be formed that completely encircle the interior of the Bladder laterally.

3.3.2 Bladder assembly construction.

3.3.2.1 Basic Bladder. The Bladder shall be fabricated from 2-ply coated fabric as specified herein. The configuration and dimensions of the Bladder and the location of ports, fittings, handles, and restraint system shall be as shown in Figure 1-A. For both Bladder interior and exterior, all coated fabric edges (seams, patches, and fabric flanges of fittings) shall be covered with coated fabric or coating compound used to fabricate the Bladder. Coated fabric panels may

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be spliced together to make up the length of the Bladder. All Bladder seams between adjacent panels shall be constructed to prevent wicking through the nylon fabric.

3.3.2.2 Forward restraint harness. The Forward Restraint Harness shall be built into the forward section of the Bladder as shown in Figure 4-A. The harness shall consist of fourteen (14) straps each twenty-four (24) foot which wrap around the front of the Bladder and are bonded to the Bladder. Each individual strap shall attach to a "D" Ring in a fashion to enable the "D" to be an integral part of the forward restraint harness and Bladder. This shall be accomplished on both ends of the strap. The underside straps are attached to a twenty-seven (27) inch fixed length strap (not part of the Bladder) to the appropriate tie-down on the platform utilized for transportation. The topside straps are attached to a sixty (60) inch adjustable length strap (not part of the Bladder) to the appropriate tie-down on the platform. The straps shall be positioned on the Bladder such that they run diagonally across the Bladder, around the nose, and under the Bladder. All fourteen (14) straps shall be configured so that the longest length of strap on the top surface results in the shortest length of strap underneath and vice versa. This technique of diagonal straps and bonding results in the Bladder being constrained by twenty-eight (28) attachments to the Army 11-1-2780 Type V Aerial Delivery Platform in addition to the twenty (20) attachments for the lateral strapping which utilize the ten (10) double D rings.

3.3.2.3 Lift handles. The lift handles shall be approximately 1.75 to 2.0 inches wide U-shaped straps. They shall be fastened to a coated fabric patch similar to the Bladder material. The patch shall be bonded to the Bladder below the peripheral fold line. The patch shall not be placed over the Bladder's seams. There shall be eight (8) lift handles on the Bladder. The lift handles shall be placed in accordance with Figure l-A.

3.3.2.4 Fill/evacuate and vent flanges. The fill/evacuate and vent flanges shall be bonded to the coated fabric of the Bladder as shown in Figure 1-A. The manufacture shall be as shown in MIL-T-52983 Figure 6a and Figure 6b for the fill/evacuate flange and MIL-T-52983 Figure 10a and Figure 10b for the vent flange. A patch-type procedure may be utilized. The fill/evacuate and vent flanges shall have protective coatings. The minimum coating thickness shall be 0.0004 inch for castings and 0.0007 inch for wrought aluminum. All aluminum parts exposed to view on the outside of the Bladder shall have a dyed coating approximating the Bladder exterior color and shall be non-reflective.

3.3.2.5 Closure plates - fill/evacuate and vent. The closure plates shall be bolted to the fill/evacuate and vent flanges prior to shipment. The manufacture shall be as shown in MIL-T-52983 Figure 7 for the closure plate outer dimensions and bolt pattern to mate with the fill/evacuate flange (3.3.2.4). Exception to MIL-T-52983 Figure 7: No centrally located 4 inch hole or bolt pattern holes are required for the closure plate. Use the MIL-T-52983 Figure 10a dimensions for the outer bolt pattern of the vent closure plate to mate with the vent flange (3.3.2.4). Closure plates shall have protective coatings. The minimum coating thickness shall be 0.0004 inch for castings and 0.0007 inch for wrought aluminum. All aluminum parts exposed to view on the outside of the Bladder shall have a dyed coating approximating the Bladder exterior color and shall be non-reflective.

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3.3.2.6 "D" rings (forward restraint harness) tow rings and double “D” rings. The double and single "D" Ring assemblies shall be capable of being shaped and welded to form one continuous ring as shown in Figure 5-A and Figure 6-A. Installation of double "D" ring assemblies shall be as specified in 3.3.2.2.

3.3.2.7 Fill/evacuate nozzle. The fill/evacuate nozzle shall be bolted to the fill/evacuate flange utilizing an O ring sealing medium. The nozzle shall be packaged/shipped with the Bladder in the same container. It shall then be installed on the Bladder fill/evacuate flange prior to use. The manufacture shall be in accordance with MIL-T-52983 Figure 7 for the fill/evacuate plate outer dimensions and bolt pattern to mate with the fill/evacuate flange (3.3.2.4). Centrally locate 6 inch hole and bolt pattern to mate with AA59326/8-5-A Type VIII Female by Flange Coupling Half 6 inch cam lock fitting. Exception to MIL-T-52983 Figure 7: Centrally located 4 inch hole and bolt pattern is deleted and replaced by the 6 inch hole and AA59326/8-5 bolt pattern described above. Alternate means of mounting the 6 inch female cam lock such as welding may be permitted with prior Government approval. A A-A-59326/11, 4 inch Dust Plug and A-A-59326/12 6 inch Male to 4 inch Female Reducer shall also be provided in the same shipping container. Aluminum alloy fill/evacuate nozzle shall have protective coatings. The minimum coating thickness shall be 0.0004 inch for castings and 0.0007 inch for wrought aluminum. All aluminum parts exposed to view on the outside of the Bladder shall have a dyed coating approximately the Bladder exterior color and shall be non-reflective.

3.3.2.8 Vent nozzle. The vent nozzle consisting of AA59326/4-4-A Type IV Male by Flange Coupling Half 4 inch cam lock fitting shall be bolted to the vent flange utilizing an O ring sealing medium. The nozzle shall be packaged separately and shipped with the Bladder in the same container. It shall then be installed on the Bladder vent flange prior to use. Alternate means of mounting the 4 inch male cam lock such as welding may be permitted with prior Government approval. A 4 inch A-A-59326/10 Dust Cap shall be provided. The minimum coating thickness shall be 0.0004 inch for castings and 0.0007 inch for wrought aluminum. All aluminum parts exposed to view on the outside of the Bladder shall have a dyed coating approximately the Bladder exterior color and shall be non-reflective.

3.3.2.9 Tow rings. There are tow rings attached to the Bladder at the forward and aft ends as shown in Figure 4-A. The rings are located on the longitudinal centerline and are attached to the Bladder tank by NBR/nylon strips which are bonded to the Bladder tank. Figure 6-A shows the tow ring detail. The bond of the tow rings to the Bladder is to be verified in accordance with 4.5.8.13

3.3.3 Identification and information plates. The Bladder shall be identified in accordance with MIL-STD-130 by means of an identification label. The label shall be made of coated fabric and bonded to the Bladder in a readily accessible location. Additionally it shall be marked with an Item Unique Identification (IUID). UID information shall be included as both a bar code and human readable markings. The following information shall be molded, either recessed or in relief, using 0.50 inch lettering on the identification label:

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BLADDER, FABRIC
3000 U.S. GALLONS FUEL
PART NUMBER: (Specify)
NSN: (Specify)
MANUFACTURER: (Specify)
CAGE CODE: (Specify)
DATE MANUFACTURED: (Specify month (3 letters) and year (4 numbers))
CONTRACT NO: (Specify)
LOT & SERIAL NO: (Specify)
WEIGHT EMPTY: (Specify approximate weight in pounds)
DATE OF WARRANTY EXPIRATION:
(Specify month (3 letters) and year (4 numbers))
SHELF LIFE: (Specify month (3 letters) and year (4 numbers))

3.3.4 Environment, Safety, and Occupational Health (ESOH).

3.3.4.1 System safety. The design of the Bladder shall not contain any system safety mishap risk categories greater than medium as defined in Table III of MIL-STD-882.

3.3.4.2 Hazardous material. The design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The Bladder shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. The Bladder shall not contain or use hexavalent chromium. The Bladder shall not contain or use cadmium without written approval by the procuring activity. For the purposes of this requirement, the Class I ODS and controlled substances identified in AFI 32-7086 shall not be used in any system, component, or process.

3.3.4.3 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features, which may cause injury to personnel.

3.3.4.4 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the Bladder with wire ropes or chains. "Dog tag" style beaded chains shall not be provided.

3.3.4.5 Electrostatic discharge (ESD). The design of the Bladder shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking.

3.3.5 Human systems integration. The Bladder shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and Mission-Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear. Chemical Warfare Gear is not required for preventive maintenance or major corrective maintenance.

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3.3.6 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components. Tapped threads shall have a minimum thread engagement in accordance with Table I.

TABLE I. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

3.3.7 Service and Storage life.

3.3.7.1 Service life temperature range. When withdrawn from storage for operational use, the Bladder and components shall be suitable for operational use, for a minimum of one (1) year at ambient temperatures from -20 °F to 140 °F. The Bladder, secured to a transportation platform in either the filled or the unfilled state, shall not leak under normal continued operational use when tested in accordance with 4.5.6.1.

3.3.7.2 Storage life temperature range. The Bladder and associated components shall withstand folded storage in a shipping container for a minimum of five (5) years at ambient temperatures from -65 °F to 160 °F without damage or leakage, when subsequently filled with fuel and tested in accordance with 4.5.6.2.

3.3.7.3 Hydrocarbon fuel contact. The Bladder shall be capable of use with continuous contact with hydrocarbon fuels with 40% aromatics, rainwater, and water associated with fuels when tested in accordance with 4.5.6.4.

3.3.7.4 Air leakage. The Bladder shall be capable of withstanding an internal air pressure of 2.25 pounds per square inch gauge (PSIG) evidence of leakage when tested in accordance with 4.5.6.5.

3.3.7.5 Fuel pressure leakage test. The Bladder shall withstand a pressure, measured by gauge, of 150% (2.25 PSIG) of the normal pressure encountered during service for the duration of 15 minutes without evidence of leakage. The Bladder shall not leak under normal transportation and use when tested in accordance with 4.5.6.6.

3.3.7.6 Surge damping. The Bladder shall dampen the fluid surge and not allow self-amplification of the fluid surge during normal operational use when tested in accordance with 4.5.6.7.

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3.3.7.7 Load stresses. In the filled condition, the Bladder and its integral restraints shall withstand a forward stress load of 3.0 g's, an aft load of 1.5 G's, a vertical load of 2.0 G's, and a transverse load of 1.5 G's. The Bladder shall withstand an 8.0 g forward stress (overload) condition when tested in accordance with 4.5.6.8.

3.3.7.8 Fill/vent rate. The Bladder, with fill/evacuate nozzle and vent nozzle installed, shall permit fill/evacuate rates of 600 gallons per minute when tested in accordance with 4.5.6.9.

3.4 Environmental conditions.

3.4.1 Operating temperature range. The Bladder shall be capable of operating in ambient temperatures ranging from -20 °F to 140 °F when tested in accordance with 4.5.7.1.

3.4.2 Storage temperature range. The Bladder shall be capable of being stored in ambient temperatures ranging from -65 °F to 160 °F when tested in accordance with 4.5.7.1.

3.4.3 Humidity. The Bladder shall show no degradation from humidity exposure either visually or hydrostatically when tested in accordance with 4.5.7.2.

3.4.4 Vibration. The shipping container shall show no breakage, loosening of skids, bolt distortion, nor any anomaly that may affect service or storage life. The Bladder shall show no deformation, seam damage, cracking or peeling, separation of bonds, chafing, or any anomaly that may affect the mission or service life. Both these requirements shall be met when tested in accordance with 4.5.7.3.

3.4.5 Salt-fog. The Bladder shall show no degradation from salt-fog exposure either visually in accordance with 4.5.1 or hydrostatically when tested in accordance with 4.5.7.4.

3.4.6 Fungus. All materials used in the Bladder shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The Bladder shall be suitable for operation and storage in conditions encountered in a tropical environment. The Bladder shall show no ability to support spore growth when tested in accordance with 4.5.7.5.

3.4.7 Ozone resistance. The Bladder shall show no evidence of cracking or checking after exposure to an ozone environment when tested is accordance with 4.5.7.6.

3.4.8 Solar radiation. The Bladder shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments and high altitudes.

3.5 Component Requirements.

3.5.1 Lift handle. When inspected in accordance with 4.5.8.1, the bonds between each handle patch and the Bladder fabric shall be capable of withstanding perpendicular loads of 1000 pounds.

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3.5.2 Forward restraint harness. When inspected in accordance with 4.5.8.2, the restraint harness with the attached "D" ring shall be capable of withstanding a 10,000-pound tensile load without breakage of the nylon weave or wraparound attachment of the "D" ring. The stitching of the sewn areas for the harness-to-harness overlay attachments shall show no evidence of degradation. Total elongation of the harness under the tensile load shall not exceed 25% nor 10% when load is relaxed.

3.5.3 Nylon tie cord. When inspected in accordance with 4.5.8.3, the nylon tie cord shall be capable of withstanding a 450-pound tensile load without breakage of external woven overlay or seven (7) internal cords. Elongation shall not exceed 25% under tensile load or 10% when load is relaxed.

3.5.4 Fluid flow restrictor tie-down. When inspected in accordance with 4.5.8.4, the fluid flow restrictor tie down shall be subjected to a tensile load of 1500 pounds through the eyelets. There shall be no degradation of the bond and no eyelet tearing through the cloth member. Distortion of the eyelet is permissible where cable contact is made.

3.5.5 Bladder material - tensile load. When inspected in accordance with 4.5.8.5, the Bladder material sample shall be subjected to a 3000-pound tensile load. The sample shall not show evidence of degradation in the form of tearing or breaking of the fabric weave, no separation of weave to coating, and no cracks or tears in the coating. The sample elongation shall not exceed 25% under load nor 10% when load is relaxed.

3.5.6 Bonded seam-in-line tensile load. When inspected in accordance with 4.5.8.6, the seam shall show no evidence of degradation in the form of bonding, cracking, fabric tear, sealing or separation of weave or coating on either side of the Bladder.

3.5.7 Bonded seam-across-seam tensile load. When inspected in accordance with 4.5.8.7, the seam shall show no evidence of degradation in the form of bonding, cracking, fabric tear, sealing or separation of weave or coating on either side of the Bladder.

3.5.8 Bonded fittings (vent-fill/evacuate flanges). When inspected in accordance with 4.5.8.8, the fittings shall show no evidence of separation of the bond between the flange and the Bladder skin.

3.5.9 "D" rings - tensile load. When inspected in accordance with 4.5.8.9, the "D" rings shall have a 10,000 tensile load applied in the direction from the base to the nose. After this load has been relaxed, the "D" rings shall show no evidence of breakage or distortion.

3.5.10 "D" rings - metallurgy. When inspected in accordance with 4.5.8.10, the inspection results shall indicate that the material is equivalent to ASTM-A276 Type 304L Stainless Steel.

3.5.11 Nozzle assemblies - metallurgy. When inspected in accordance with 4.5.8.11, the inspection results shall indicate that the material is equivalent to ASTM-B221 Type 6061-T6 Aluminum.

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3.5.12 Vent and fill/evacuate flanges - metallurgy. When inspected in accordance with 4.5.8.12, the inspection results shall indicate that the material is equivalent to ASTM-B221 Type 6061-T6 Aluminum.

3.6 Workmanship. Workmanship shall be the highest quality and shall permit no defects adversely affecting the strength or serviceability of the finished Bladder. The reinforced flange type fittings shall contain no voids or cracks that could adversely affect the strength of the assembly. All metal parts shall be free of sand, dirt, and scale. Surfaces shall be smooth with edges rounded or beveled. The inside and the outside of the Bladder shall be clean and free of all foreign material. Any necessary repair and rework shall restore the reworked area to its full strength and shall meet all the applicable requirements of this specification. The bonded surfaces of all seamed areas, fitting flanges, and patch-type repairs shall affect a bond that will result in the strength of the bonded area not less than the strength of the adjacent Bladder fabric.

3.6.1 Repair and rework of Bladders. Any repair and rework of any Bladder shall be performed prior to quality conformance inspection with the exception of air leakage. Repair and rework of a seam shall not exceed 10% percent of the length of the seam. Repair and rework of panels shall be limited to areas of damage not greater than six (6) inches in length, six (6) inches in width, or two (2) inches in diameter. All repair and rework procedures shall be in accordance with the Repair and Rework Plan.

3.6.2 Repair and rework plan. The contractor shall prepare a Repair and Rework Plan. The plan shall include a specific description of methods and procedures for the repairs and/or rework of anomalies, which might occur during the manufacturing or assembly of the Bladder. The plan shall include documented work instructions which shall be used by the contractor to ensure that repaired and reworked areas have structural integrity equal to or greater than seams or panels that have not been reworked. No part of this plan shall conflict with or take precedence over the requirements of this specification. The plan shall be made available, upon request, to the Government.

3.6.3 Special tools. The design of the Bladder shall minimize the requirement for special tools (see 6.4.11). All special tools shall be provided with, and stored with, the Bladder.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

Requirements shall be verified in accordance with Table II.

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TABLE II. Requirement verification matrix.

Section 3 Requirement	Verification Method	Section 4 Verification
3.1 First article inspection.	Not Applicable (N/A)	N/A
3.1.1 Bladders for testing.	Examination	4.5.1 Examination of product.
3.2 Materials.	Examination	4.5.1 Examination of product.
3.2.1 Material deterioration, prevention, and controls.	Examination	4.5.1 Examination of product.
3.2.1.1 Dissimilar metals.	Examination	4.5.1 Examination of product.
3.2.1.2 Identification of materials and finishes.	Examination	4.5.1 Examination of product.
3.2.1.3 Recycled, recovered, environmentally preferable, or biobased materials	Examination	4.5.1 Examination of product.
3.2.2 Coated fabric.	Examination	4.5.1 Examination of product.
3.2.2.1 Coating compounds.	Examination	4.5.1 Examination of product.
3.2.2.2 Nylon cloth.	Examination	4.5.1 Examination of product.
3.2.2.3 Weathering resistance.	Examination	4.5.1 Examination of product.
3.2.3 Fill/evacuate and vent nozzle material.	Examination	4.5.1 Examination of product.
3.2.4 Fill/evacuate and vent flange material.	Examination	4.5.1 Examination of product.
3.2.5 "D" rings material.	Examination	4.5.1 Examination of product.
3.3 Design and construction.	Examination	4.5.1 Examination of product.
3.3.1 Bladder and associated component design.	N/A	N/A
3.3.1.1. Basic Bladder.	Examination	4.5.1 Examination of product. 4.5.2.1 Basic Bladder.
3.3.1.2 Double “D” tie down rings.	Examination	4.5.1 Examination of product.
3.3.1.3 Lift handles.	Examination	4.5.1 Examination of product.
3.3.1.4 Fill/evacuate assembly.	Examination	4.5.1 Examination of product.
3.3.1.5 Vent assembly.	Examination	4.5.1 Examination of product.
3.3.1.6 Reed valve vapor eliminator.	Examination Test	4.5.1 Examination of product. 4.5.6.9 Fill/Vent Rate.
3.3.1.7 Fluid flow restrictors.	Examination	4.5.1 Examination of product.
3.3.2 Bladder assembly construction.	N/A	N/A
3.3.2.1 Basic Bladder.	Examination	4.5.1 Examination of product. 4.5.2.1.1 Basic Bladder.
3.3.2.2 Forward restraint harness.	Examination	4.5.1 Examination of product. 4.5.2.1.2 Forward restraint harness.
3.3.2.3 Lift handles.	Examination	4.5.1 Examination of product. 4.5.2.1.3 Lift handles.

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TABLE II. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.3.2.4 Fill/evacuate and vent flanges.	Examination	4.5.1 Examination of product. 4.5.2.1.4 Fill/evacuate and vent flanges.
3.3.2.5 Closure plates - fill/evacuate and vent flange.	Examination	4.5.1 Examination of product. 4.5.2.1.5 Closure plate - fill/evacuate and vent flange.
3.3.2.6 "D" rings (forward restraint harness) and double “D” rings.	Examination	4.5.1 Examination of product. 4.5.2.1.6 Double “D" tie down rings. 4.5.2.7 "D" rings (forward restraint harness and tow rings).
3.3.2.7 Fill/evacuate nozzle.	Examination	4.5.1 Examination of product. 4.5.2.1.8 Fill/evacuate nozzle.
3.3.2.8 Vent nozzle.	Examination	4.5.1 Examination of product. 4.5.2.1.9 Vent nozzle assembly.
3.3.2.9 Tow rings.	Examination Test	4.5.1 Examination of product. 4.5.8.13 Tow Ring tensile load.
3.3.3 Identification and information plates.	Examination	4.5.1 Examination of product.
3.3.4 Environment, Safety, and Occupational Health (ESOH).	N/A	N/A
3.3.4.1 System safety.	Analysis	4.5.3 System safety hazard analysis.
3.3.4.2 Hazardous material.	Examination	4.5.1 Examination of product.
3.3.4.3 Component protection.	Examination	4.5.1 Examination of product.
3.3.4.4 Foreign object damage (FOD).	Examination	4.5.1 Examination of product.
3.3.4.5 Electrostatic discharge (ESD).	Analysis	4.5.4 Electrostatic discharge analysis.
3.3.5 Human systems integration.	Demonstration	4.5.1 Examination of product.
3.3.6 Fastening devices.	Examination	4.5.1 Examination of product.
3.3.7 Service and storage life.	N/A	
3.3.7.1 Service life temperature range.	Analysis Test	4.5.5.1 Service life analysis. 4.5.6.1 Service life.
3.3.7.2 Storage life temperature range.	Analysis Test Test	4.5.5.2 Storage life analysis. 4.5.6.2 Storage life. 4.5.6.3 Storage life temperature.
3.3.7.3 Hydrocarbon fuel contact.	Test	4.5.6.4 Hydrocarbon fuel contact.
3.3.7.4 Air leakage.	Test	4.5.6.5 Air leakage.
3.3.7.5 Fuel pressure leakage test.	Test	4.5.6.6 Fuel pressure leakage test.

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TABLE II. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.3.7.6 Surge damping.	Analysis	4.5.6.7 Surge damping.
3.3.7.7 Load stresses.	Analysis	4.5.6.8 Load stresses.
3.3.7.8 Fill/vent rate.	Test	4.5.6.9 Fill/vent rate.
3.4 Environmental conditions.	N/A	4.5.7 Environmental testing.
3.4.1 Operating temperature range.	Test	4.5.7.1 Operating and storage temperature tests.
3.4.2 Storage temperature range.	Test	4.5.7.1 Operating and storage temperature tests.
3.4.3 Humidity.	Test	4.5.7.2 Humidity test.
3.4.4 Vibration.	Test	4.5.7.3 Vibration test.
3.4.5 Salt-fog.	Test	4.5.7.8 Salt-fog test.
3.4.6 Fungus.	Test	4.5.7.9 Fungus test.
3.4.7 Ozone resistance.	Test	4.5.7.10 Ozone resistance test.
3.4.8 Solar radiation.	Examination	4.5.1 Examination of product.
3.5 Component requirements.	N/A	N/A
3.5.1 Lift handle.	Test	4.5.8.1 Lift handle.
3.5.2 Forward restraint harness.	Test	4.5.8.2 Forward restraint harness.
3.5.3 Nylon tie cord.	Test	4.5.8.3 Nylon tie cord.
3.5.4 Fluid flow restrictor tie-down.	Test	4.5.8.4 Fluid flow restrictor tie-down.
3.5.5 Bladder material - tensile load.	Test	4.5.8.5 Bladder material-tensile load.
3.5.6 Bonded seam-in-line tensile load.	Test	4.5.8.6 Bonded seam - in-line tensile load.
3.5.7 Bonded seam-across-seam tensile load.	Test	4.5.8.7 Bonded seam - across seam tensile load.
3.5.8 Bonded fittings (vent-fill/evacuate flanges).	Test	4.5.8.8 Bonded fittings (vent-fill/evacuate flanges).
3.5.9 "D" rings - tensile load.	Test	4.5.8.9 "D" rings - tensile load.
3.5.10 "D" rings - metallurgy.	Test	4.5.8.10 "D" rings - metallurgy.
3.5.11 Nozzle assemblies - metallurgy.	Test	4.5.8.11 Nozzle assemblies - metallurgy.
3.5.12 Vent and fill/evacuate flanges - metallurgy.	Test	4.5.8.12 Vent and fill/evacuate flanges - metallurgy.
3.6 Workmanship.	Examination	4.5.1 Examination of product.
3.6.1 Repair and rework of Bladders.	Examination	4.5.1 Examination of product.
3.6.2 Repair and rework plan.	Examination	4.5.1 Examination of product.
3.6.3 Special tools.	Examination	4.5.1 Examination of product.

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4.2 First article inspection. The first article Bladder(s) shall be subjected to the analyses, demonstrations, and examinations of 4.5, and tests described in 4.5.6 through 4.5.8. Presence of one or more defects shall be cause for rejection. The contractor shall provide or arrange for all test equipment and facilities. Unless otherwise approved by the procuring activity all first article Bladders shall be in the same configuration at all times and configuration changes shall not be made during the first article Bladder inspection. A first article inspection shall be performed on one (1) Bladder when a first article sample is required (see 3.1 and 6.2).

4.3 Conformance inspection. Each production Bladder shall be subjected to the examination described in 4.5.1. Conformance inspection shall ensure that production items meet specification requirements prior to acceptance by the Government. Each Bladder shall be inspected for compliance with the requirements specified in 3.3.2 through 3.3.6, and 3.3.7.4. Any redesign or modification of the product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention to adequacy and suitability. This element of inspection shall encompass all visual, dimensional, and performance requirements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Inspection requirements.

4.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type and calibrated at proper intervals to ensure accuracy.

4.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

Test rejection criteria. Throughout all tests specified herein, the Bladder shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

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c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturers (OEM's) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

4.5 Test methods.

4.5.1 Examination of product. Each Bladder shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 4.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each Bladder function shall be verified. Certifications and analyses shall be provided in accordance with Table III. Each production Bladder shall be inspected to a Government-approved reduced version of the checklist.

TABLE III. Certifications and analyses.

Paragraph	Required Certifications and Analyses
3.3.4.1 System safety.	Contractor system safety hazard analysis (see 4.5.3).
3.3.4.5 Electrostatic discharge.	Contractor analysis of the electrostatic discharge requirement (see 4.5.4).
3.3.7.1 Service life temperature range.	Contractor analysis of the service life requirement (see 4.5.5.1).
3.3.7.2 Storage life temperature range.	Contractor analysis of the service life requirement (see 4.5.5.2).
3.3.7.6 Surge damping.	Contractor analysis of the surge damping requirement (see 4.5.6.7)
3.3.7.7 Load stresses.	Contractor analysis of the load stresses (see 4.5.6.8).

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TABLE III. Certifications and analyses – Continued.

Paragraph	Required Certifications and Analyses
3.4.6 Fungus.	Contractor certification that the materials used in construction of the Bladder are fungus resistant or suitably treated to resist fungus.
3.4.8 Solar radiation.	Contractor certification that the Bladder performance is not adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

4.5.1.1 Examination setup. The Bladder shall be laid out flat in an inspection area that allows access from all sides. The area shall be large enough for the Bladder to be pulled from one end back over itself enabling the Bladder to be turned upside down. The minimum required floor space of the inspection area is 12 feet wide by 40 feet long.

4.5.1.2 Examination procedure. The following visual examinations shall be performed and the results thereof recorded on the appropriate data sheets produced by contractor.

4.5.1.2.1 Bladder assembly. The entire external surface shall be visually examined for continuity, uniformity, surface texture, bonding of seams, patches, workmanship, appearance, and any anomaly that may affect service use. In addition, lift handle location and count shall be made along with the location and total of lateral tie down Double "D" Rings. The Double "D" Rings shall be located on the transverse center-line longitudinally along the top side of the Bladder.

4.5.1.2.2 Harness layout and attachment configuration. The forward restraint harness layout and attachment configuration on top of the Bladder shall be uniform and equal on both lateral sides of the Bladder. The forward attachment technique and patch shall be conformable and uniform across the forward end of the Bladder. The Bottom Harness Configuration shall not be bonded to the Bladder behind the forward attachment point (see 3.3.2.2).

4.5.1.2.3 Lift handles. The lift handles shall be visually examined for uniformity of casting or machining, workmanship, freedom from burrs and sharp edges, cracks, protective coating, and any anomaly that may affect service use (see 3.3.2.3).

4.5.1.2.4 Fill/evacuate and vent flanges. The fill/evacuate and vent flanges shall be visually examined for uniformity of casting or machining, workmanship, freedom from burrs and sharp edges, cracks, protective coating, and any anomaly that may affect service use (see 3.3.2.4).

4.5.1.2.5 Fill/evacuate and vent flange material. The closure plates shall be visually examined for uniformity of casting or machining, workmanship, freedom from burrs and sharp edges, cracks, protective coating, and any anomaly that may affect service use (see 3.3.2.5).

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4.5.1.2.6 "D" rings material. The "D" rings and double "D" rings shall be visually examined for uniformity of casting or machining, workmanship, cracks, freedom from burrs and sharp edges, and any anomaly that may affect service use (see 3.3.2.6).

4.5.1.2.7 Fill/evacuate and vent nozzle material. The fill/evacuate nozzle shall be visually examined for uniformity of casting or machining, workmanship, cracks, protective coating, freedom from burrs and sharp edges, and any anomaly that may affect service use (see 3.3.2.7).

4.5.1.2.8 Vent nozzle. The vent nozzle shall be visually examined for uniformity of casting or machining, workmanship, cracks, protective coating, freedom from burrs and sharp edges, and any anomaly that may affect service use (see 3.3.2.8).

4.5.2 Dimensional inspection. The following items shall be dimensionally examined in accordance with the appropriate Figures of this specification, to ensure compatibility with the existing system, drawing package compliance, and product assurance. The results of this inspection showing dimensions greater than allowable specified tolerances shall be considered a failure of the test.

4.5.2.1 Test procedure. The following dimensional examinations shall be performed and the results thereof recorded. Traceable measurement instrumentation shall be utilized and recorded.

4.5.2.1.1 Basic Bladder. Prior to sectioning the Bladder, the length, width, and average height shall be measured and recorded with the Bladder collapsed. The location of the lateral tie downs (Double "D" Rings), fill/evacuate and vent flanges shall be measured and recorded in accordance with Figure 1-A, (see 3.3.2.1).

4.5.2.1.2 Forward restraint harness. The forward restraint harness shall be measured for total length, width, and thickness in the assembled position. All individual harnesses shall be measured and recorded in accordance with Figure 1-A. All individual straps shall be equal to the specified length plus 2 inches minus 0 inches. The fold back of harness through "D" Ring shall be 12 inches plus 1.2 inches minus 0 inches minimum (3.3.2.2).

4.5.2.1.3 Lift handles. The lift handle shall be dimensionally examined to verify compliance with 3.3.2.3. Verify lift handle location, count, shape, and bonding technique. Two (2) lift handles shall be examined.

4.5.2.1.4 Fill/evacuate and vent flanges. The fill/evacuate flange shall be dimensionally examined to verify compliance with 3.3.2.4. Verify depth of threaded holes, diameters, thickness, thread size, shape, seal groove, and threaded hole placement.

4.5.2.1.5 Closure plate - fill/evacuate and vent flange. The vent flange shall be dimensionally examined to verify compliance with 3.3.2.5. Verify the depth of the threaded holes, diameters, thickness, thread size, shape, seal groove, and threaded hole placement.

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4.5.2.1.6 Double “D” tie down rings. The Double "D" Rings shall be dimensionally examined to verify compliance with Figure 5-A. Verify diameters, shape, angles, weldment and overall length, width, and height. Two (2) ring assemblies shall be inspected.

4.5.2.1.7 "D" rings (forward restraint harness) and tow rings. The "D" Rings shall be dimensionally examined to verify compliance with Figure 6-A. Verify diameters, shape, angles, weldment and overall length, width, and height. Two (2) rings shall be examined.

4.5.2.1.8 Fill/evacuate nozzle. The fill/evacuate nozzle shall be dimensionally examined to verify compliance with 3.3.2.7. Verify hole placement, diameters, thickness, height, shape, cover fit, locking levers, and lanyard attachment technique.

4.5.2.1.9 Vent nozzle assembly. The vent nozzle shall be dimensionally examined to verify compliance with 3.3.2.8. Verify hole placement, diameters, thickness, height, shape, cover fit, locking levers, and lanyard attachment technique.

4.5.3 System safety hazard analysis. A system safety hazard analysis of the Bladder shall be conducted in accordance with 4.3.1 through 4.3.8 of MIL-STD-882 to demonstrate compliance with the mishap risk requirement of 3.3.4.1.

4.5.4 Electrostatic discharge analysis. An engineering analysis shall be performed to demonstrate compliance with the electrostatic discharge requirement of 3.3.4.5.

4.5.5 Service and storage life analysis.

4.5.5.1 Service life analysis. An engineering analysis shall be performed to demonstrate compliance with the service life requirement of 3.3.7.1.

4.5.5.2 Storage life analysis. An engineering analysis shall be performed to demonstrate compliance with the service life requirement of 3.3.7.2.

4.5.6 Performance inspection. The Bladder shall be subjected to the following performance requirements as specified in 3.4.1 through 3.4.8.

4.5.6.1 Service life. Test in accordance with 4.5.7, Environmental testing.

4.5.6.2 Storage life. Test in accordance with 4.5.7, Environmental testing.

4.5.6.3 Storage temperature. Test in accordance with 4.5.7, Environmental testing.

4.5.6.4 Hydrocarbon fuel contact. The hydrocarbon fuel contact test shall be performed on the Bladder to verify the integrity of the Bladder when filled with fuel. Any evidence of leakage or seepage shall constitute failure of this test (see 3.3.7.3). Fill the tank with its rated capacity of diesel fuel in accordance with ASTM D975 or JP-8 in accordance with MIL-PRF-83133 or JET-A in accordance with ASTM D1655, and allow to stand for 60 days, +2 days, -0 days. During the test and at the end of this period, examine the tank for seepage and leakage. Immediately after the

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tank is emptied, the tank bottom shall be examined for leakage and seepage. The Bladder shall be fully assembled with nozzles attached and locked. The tank shall be placed outdoors without any environmental protective covering.

4.5.6.5 Air leakage. The air leakage test shall be performed on the Bladder secured to a simulated platform to verify the integrity of the Bladder fabric, fittings, and external seams when pressurized with air. Using a calibrated air pressure gauge, test/block-off plates (one with inflation valve installed), air supply, phenolphthalein crystals, denatured alcohol, water and white cloth. Attach test plates to all except one fitting. Mix phenolphthalein solution by adding 40 grams of phenolphthalein crystals to ½ gallon of ethyl alcohol and mixing until crystals have dissolved. Add ½ gallon of water to this solution. Soak an absorbent cloth with ammonia at a ratio of 1 teaspoon per cubic foot of fuel Bladder capacity. Place the saturated cloth inside the fuel Bladder and install the remaining test plate. Inflate the fuel Bladder to 2.25 PSIG maximum. Maintain pressure for 15 minutes. Soak enough white cloth to cover entire fuel cell in the phenolphthalein solution. Thoroughly wring it out and spread it smoothly on the exterior surface of the fuel Bladder. Spread the white cloth smoothly under and over the surface of the Bladder. Leaks will appear as red stains on the white cloth. Mark any leaks found and move the cloth to a new location. Once test is complete, remove cloth, test plates, and purge the fuel Bladder with fresh air. Document any signs of leakage or discrepancies. Any evidence of leakage shall constitute failure of test. The Air Leakage Test to be conducted on First Article Test unit and during production conformance testing.

4.5.6.6 Fuel pressure leakage test. Applicable to First Article Unit. The leakage test shall be performed on the Bladder secured to a simulated platform to verify the integrity of the Bladder fabric, fittings, and external seams when pressurized with fuel. Fill the Bladder with diesel or jet fuel until a total of 2400 gallons is obtained. Seal the vent nozzle by activating the locking arms. Tighten all ratchet assemblies on diagonal and lateral restraints. Continue filling the Bladder with fuel until the pressure gauge indicates 2.25 PSIG. Secure fuel and set valve controls to allow pressure gauge monitoring of the Bladder. Commence a 15-minute test time while monitoring pressure in the Bladder. At completion of the test duration, adjust valves to relieve pressure on the Bladder. The Bladder shall be fully assembled and installed on an actual or simulated Army 11-1-2780 Type V, aerial delivery platform. The Type V aerial delivery system shall be simulated by laying out on a 108 inch by 240 inch level surface capable of supporting the Bladder filled with fuel, two 240 inch length, 3 inch deep channel or equivalent. Along the length of the simulated Type V side rail clevises will be mounted to accept 10 lateral and 14 diagonal tie down straps. The Type V aerial delivery platform provides for up to 40 tie down points along the 20 foot length of the platform, but only 24 are used. The spacing of the tie down points repeats a pattern each 2 feet beginning from the first edge. The first clevis location is 1.7 inch followed by 6.875 inch on center followed by 6.875 inch followed by 6.875 inch. At multiples of 2 feet from the first edge this pattern is repeated for a total of 10 times for a total of 40 tie down locations. The left and right channels shall be mounted on the previous mentioned surface spaced 108 inches apart. All diagonal restraints and lateral restraints shall be connected per the attached Figure 4-A, using ratchet straps and clevises. The vent nozzle assembly shall be left unfastened. The fill/evacuate nozzle shall be installed and sealed. A fuel source shall be connected to the fill/evacuate pressure gauge port in a configuration that fill and pressure readings may be accomplished. The fuel

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source must contain a flow meter to indicate total flow in gallons. Failure of the Bladder to maintain the specified pressure without leakage shall be cause for rejection (see 3.4.5).

4.5.6.7 Surge damping. A demonstration shall be conducted to ensure that fluid surges are damped. With the Bladder filled at both 1,500 gallon fuel and 3,000 gallon fuel and strapped to the Type V platform, or simulated platform, a surge initiated at the aft end of the Bladder will not result in a sustained oscillation of fuel (see 3.3.7.6). A surge is initiated by dropping a 100 pound minimum weight from 1 foot height minimum on top of the Bladder at the rear. No more than 3 reflected waves which displace the weight greater than 1 inch are permitted.

4.5.6.8 Load stresses. A detailed engineering evaluation shall be conducted on the tank to determine compliance with the tests specified in MIL-STD-1791 for air transport in types C-5, C-17 and C-130 aircraft (see 3.3.7.7).

4.5.6.9 Fill/vent rate. The 600-gallons per minute (GPM) flow rate into the tank shall be measured utilizing a calibrated flow meter. Evacuation rates of fuel from the tank shall be 600 GPM and be demonstrated and measured by means of a calibrated flow meter. The flow rates shall be maintained for a minimum of 2 minutes. Failure to maintain stated flow rate shall be cause for rejection (see 3.3.7.8).

4.5.6.10. First Article Testing Sequence.

1. Air Test in accordance with 4.5.6.5
2. Low Temperature Test & Visual Inspection in accordance with 4.5.7.1.1
3. High Temperature Test & Visual Inspection in accordance with 4.5.7.1.2
4. Humidity Test & Visual Inspection in accordance with 4.5.7.2
5. Vibration Test & Visual Inspection in accordance with 4.5.7.3
6. Salt Fog Test & Visual Inspection in accordance with 4.5.7.4
7. Air Test in accordance with 4.5.6.5 (repeat following tests 2 through 6)
8. Fuel Pressure Leak Test in accordance with 4.5.6.6
9. Hydrocarbon Fuel Contact Test in accordance with 4.5.6.4

4.5.6.11. Production Lot Testing Sequence.

1. Air Test in accordance with 4.5.6.5

4.5.7 Environmental testing. The Bladder shall be subjected to the environmental tests specified in 4.5.7.1 through 4.5.7.6 to verify performance of the Bladder after exposure. Environmental testing shall be done in accordance with MIL STD 810. Following the environmental testing the Bladder shall be Air leak tested in accordance with 4.5.6.5.

4.5.7.1 Operating and storage temperature tests.

4.5.7.1.1 Low temperature. The Bladder shall be subjected to low temperature in accordance with MIL-STD-810 Method 502.5, Procedure (I and II), using a chamber temperature of -40 °F. Start a 72-hour soak at -40 °F. At the completion of 72 hours, adjust the chamber ambient to

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75 °F. Maintain this condition for 24 hours. At the completion of 24 hours, remove the shipping container from the chamber and place it in an assembly area. Remove the Bladder from the shipping container. Perform a visual examination in accordance with 4.5.1 if it can be stored, manipulated, and operated under pertinent low temperature conditions without experiencing physical damage or deterioration in performance. Failure of this test shall be cause for rejection (see 3.4.1).

4.5.7.1.2 High temperature. The Bladder shall be tested in accordance with MIL-STD-810, Method 501.6, Procedure I and II, to demonstrate compliance with the high temperature storage and operating requirements of 3.4.1 and 3.4.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization. Adjust ambient air temperature to Condition 1 in Table IV. When conditions and temperature are met, continue with the 24-hour cycle in Table IV. Repeat the 24 hour cycle until a total of seven (7) cycles are completed. Return ambient temperature to 75 °F and stabilize for 24 hours. Remove the shipping container from the chamber and place it in an assembly area. Remove the Bladder from the shipping container. Perform a visual examination in accordance with 4.5.1.

TABLE IV. High temperature 24-Hour cycle.

TIME (Hours)	TEMP (°F)	TIME (Hours)	TEMP (°F)
0000 Cond 1	95	1300	156
0100	95	1400	158
0200	94	1500	160
0300	94	1600	158
0400	92	1700	153
0500	92	1800	145
0600	91	1900	131
0700	97	2000	118
0800	104	2100	105
0900	111	2200	103
1000	124	2300	99
1100	133	2400	95
1200	145		

4.5.7.2 Humidity test. The Bladder shall be subjected to a humidity test in accordance with MIL-STD-810 Method 507.5, Procedure II, using the cyclic high humidity cycle to demonstrate compliance with the requirements of 3.4.3. The Bladder shall be installed in a test chamber on a non-conductive, nonferrous platform. The vent and fill/evacuate caps shall be installed and locked. The Bladder shall be pressurized with air to 0.25 PSIG through the fill/evacuate nozzle pressure port. Gradually raise the internal chamber temperature to 140 °F and relative humidity (RH) to 95% ±5% over a period of two (2) hours. The rate of temperature change shall not exceed 1 °F per minute. Maintain these conditions for six (6) hours. Reduce the chamber ambient temperature to 86 °F over an eight (8) hour period, while maintaining the RH at 85% or greater. Raise RH to 95% ±5%. Maintain 86 °F at RH 95% ±5% for eight (8) hours. Repeat the above cycle for an additional nine (9) times for 240 hours of exposure. At the completion of 240 hours of exposure,

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return the chamber ambient to room. With the Bladder inflated, perform a visual examination. Deflate the Bladder and remove it from the-test chamber. Set up the Bladder in the assembly area and perform a visual examination in accordance with 4.5.1. Failure of this test shall be cause for rejection (see 3.4.3).

4.5.7.3 Vibration test. The Bladder shall be subjected to the vibration test to determine the resistance of the Bladder to vibration stresses expected in its shipment. Failure of this test shall be cause for rejection (see 3.4.4). The Bladder, in its shipping container, shall be mounted on a vibrator. Securing of the container shall be achieved by the use of ratchet type tie down straps used by the common carrier. The straps shall be equidistantly placed transversely on the longitudinal axis. Expose the Bladder and shipping container to sinusoidal vibration from 10 Hz to 500 Hz at the power spectral density levels required for axis direction as per MIL-STD-810. Exposure duration shall be two (2) hours per axis. Axis sequence shall be vertical, transverse, and longitudinal. At the completion of six (6) hours of vibration, the shipping container with the Bladder shall be moved to an assembly area. A visual examination of the shipping container and Bladder shall be performed in accordance with 4.5.1.

4.5.7.4 Salt-fog test. The tank assembly shall be subjected to salt fog in accordance with MIL-STD-810 Method 509.5, to demonstrate compliance with 3.4.5. Test duration shall be alternating 24-hour periods of salt fog exposure and drying conditions for 24-hour periods (two wet and two dry). The Bladder shall be installed in a test chamber on a non-conductive, nonferrous platform. The vent and fill/evacuate caps shall be installed and locked. The Bladder shall be pressurized with air to 0.25 PSIG through the fill/evacuate nozzle pressure gauge port. The Bladder shall be tested in accordance with MIL STD 810, Method 509.6, to demonstrate compliance with 3.3.6. Test duration shall be alternating 24-hour periods of salt fog exposure and drying conditions for 24-hour periods (two wet and two dry). The Bladder shall be washed with tap water to remove salt deposits to aid in the visual examination. At the completion of the visual examination, deflate the Bladder and set it up in an assembly area. Failure of this test shall be cause for rejection (see 3.4.5).

4.5.7.5 Fungus test. Fabric samples adequately representing applicable portions of the Bladder that might be damaged by exposure to fungus attack shall be subjected to a fungus test in accordance with MIL-STD-810 Method 508.6, using test duration of 28 days to demonstrate compliance with the requirements of 3.4.6. Three (3) sample fabric parts and a component sample as described in 4.5.7.1 through 4.5.7.4 shall be obtained from the Bladder utilized for destructive testing as specified in 4.2. The test samples shall be tested in accordance with MIL-STD-810, Method 508.7. Cleaning and preparation for inoculation shall be performed in accordance with MIL-STD-810, Method 508.7. The test shall be interrupted at seven (7) day intervals, during a total of 28 days, to ascertain spore growth on control samples.

4.5.7.5.1 Sample 1. Sample 1 shall be a four (4) inch by four (4) inch piece of five-layer sandwich material.

4.5.7.5.2 Sample 2. Sample 2 shall be a three (3) inch by six (6) inch piece of forward restraint harness.

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4.5.7.5.3 Sample 3. Sample 3 shall be a four (4) inch by four (4) inch piece of fluid flow restrictor material.

4.5.7.5.4 Sample 4. Sample 4 shall be one (1) vent nozzle.

4.5.7.6 Ozone resistance test. The Bladder shall be subjected to an ozone resistance test to assess the estimation of the resistance of the Bladder compound to an atmosphere containing ozone. Three (3) test samples as described in 4.5.7.6.1 through 4.5.7.6.3 shall be prepared. Test apparatus shall be of a material that will not alter test results by reaction to ozone atmosphere. The test samples shall be placed in an ozone-free atmosphere for a period of 24 hours and then placed in the test chamber. The chamber ambient temperature shall be adjusted to 104 ±4 °F and to an exposure of ozone concentration in air at a partial pressure of 50 mPa. The air-ozone velocity in the chamber shall be at least two (2) feet per second. The samples shall be exposed to these conditions for a period of 14 days. At the completion of testing with specimens in mounting fixtures, a visual inspection with 7X magnification shall be performed. Evidence of cracking or checking after exposure to ozone shall be cause for rejection (see 3.4.7).

4.5.7.6.1 Sample 1. Sample 1 shall be a one (1) inch by six (6) inch strip configured such that a 20% percent elongation load is applied. The sample will be prepared from the same rubber compound and thickness used in the bladder construction but without the nylon cloth reinforcement.

4.5.7.6.2 Sample 2. Sample 2 shall be a one (1) inch by 3.75 inch strip configured in a bent loop fashion and attached at the ends of the strip in a clamping device. The sample will be prepared from the same rubber compound and thickness used in the bladder construction but without the nylon cloth reinforcement.

4.5.7.6.3 Sample 3. Sample 3 shall be a tapered strip measuring 0.75 inch to one (1) inch in width and six (6) inches long configured in a clamping device with a 20% percent elongation load. The sample will be prepared from the same rubber compound and thickness used in the bladder construction but without the nylon cloth reinforcement.

4.5.8 Destructive testing. Destructive testing shall be performed on a completed Bladder selected from pre-production for destructive testing (see 3.1.1). The Bladder shall be sectioned for testing sample fabric parts, subsystem components, chemical analysis, workmanship, methodology, inspection, and installation technique of fluid flow restrictors. The following tests shall be performed on this Bladder and are detailed in the following list:

a. Lift handle.

b. Forward restraint harness.

c. Nylon tie cord.

d. Fluid flow restrictor tie-down.

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e. Bladder material-tensile load.

f. Bonded seam-in-line tensile load.

g. Bonded seam-across-seam tensile load.

h. Bonded fittings (vent-fill/evacuate flanges).

i. “D” rings - tensile load.

j. “D” rings – metallurgy.

k. Nozzle assemblies – metallurgy.

l. Vent-fill/evacuate flanges – metallurgy.

For the requirements of 3.5.1 through 3.5.12, the appropriate part/component shall be sectioned from the Bladder. The recommended sectioning to be followed is to cut laterally across the Bladder just behind the seam, behind the last forward restraint harness attachment point. Place a longitudinal cut to the left or right of the vent flange from the lateral cut to the rear of the Bladder stopping at the peripheral fold line. These two cuts shall enable sectioning of the majority of components required for testing.

4.5.8.1 Lift handle. The Bladder lift handle shall be subjected to a tensile load test to verify that the workmanship and bonding techniques are compliant with the requirements of this specification. The lift handle and patch assembly shall display no damage, permanent distortion, separation of the lift handle from the patch, or separation of the patch from the Bladder skin. The test sample shall consist of the lift handle and a minimum of six (6) inches of Bladder skin material around the patch area. The Bladder skin shall be clamped in a configuration to prevent movement with the lift handle centrally located within the clamps. With the sample held securely, tension shall be applied through a one (1) inch diameter bar or pipe inserted through the loop of the lift handle in a perpendicular direction. The tension shall be applied slowly and smoothly until the specified load is reached. Two (2) samples shall be tested. Apply the tension slowly to the lift handle until a load of 1500 pounds is attained. Maintain the load for duration of one (1) minute. Release the load, remove the bar and retaining clamps, and perform a visual examination. Any of the above shall constitute failure of this test (see 3.5.1).

4.5.8.2 Forward restraint harness. The Bladder forward restraint harness shall be subjected to a tensile load test to verify that the workmanship and bonding techniques are compliant with the requirements of this specification. Two (2) samples of the forward harness shall be cut from the restraint system, each being 24 inches in length excluding the "D" Ring. The samples shall be installed in a fixture capable of attachment to the “D” Ring on one end and clamping of the harness on the other. A sampling length of 12 inches shall be marked on the harness to enable elongation measurement at tensile load and load removed conditions. Apply the tensile load as uniformly as possible until a load of 10,000 pounds is attained. While maintaining the load, determine the elongation of the harness by measuring the distance between the marked areas, and record data. Remove the tensile load and again measure the distance between the marked areas to determine

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the relaxed elongation and record data. Calculate elongation for both measurements as a percentage. Perform a visual examination. Failure of the tensile load test shall be cause for rejection (see 3.5.2).

4.5.8.3 Nylon tie cord. The Bladder nylon tie cord used to fasten the fluid flow restrictor to the Bladder, shall be subjected to a tensile load test to verify that its strength and elongation is compliant with the requirements of this specification. Install a sample of nylon tie cord into a tensile test machine utilizing lockjaw mechanism to prevent cord slippage during tensile loading. Allow a 4-inch minimum exposure of cord to attach a 4-inch extensometer. Apply the tensile load uniformly until a 450-pound load is attained. Maintain the load for one (1) minute. Measure and record the elongation of the cord. Remove the tensile load and measure and record the relaxed elongation. Failure of the tensile load test shall be cause for rejection (see 3.5.3).

4.5.8.4 Fluid flow restrictor tie-down. The Bladder fluid flow restrictor tie down shall be subjected to a tensile load test to verify that the workmanship, bonding technique, and strength of eyelets are compliant with the requirements of this specification. Remove from the destructive test Bladder one (1) fluid flow restrictor tie down allowing six (6) inches of material in all directions from the bonded area. Clamp the Bladder material to a load frame in a manner to allow a vertical pull on the fluid flow restrictor so that only the bonded area is tested. Loop a cable through each eyelet and around the spreader bar to allow equal loading to the eyelets. Attach the spreader bar to the loading device. Apply a tensile load uniformly until the 1500-pound load is attained. Maintain the load for one (1) minute duration. Remove the load and perform a visual examination of the bonded area. Failure of the tensile load test shall be cause for rejection (see 3.5.4).

4.5.8.5 Bladder material-tensile load. A sample of the Bladder material shall be subjected to a tensile load test to verify that the workmanship and bonding techniques are compliant with the requirements of this specification. Cut a two (2) inch by 40-inch test sample of a five-layer sandwich section of the Bladder, from the Bladder utilized for destructive testing as specified in 4.7. Install the sample in a tensile test machine utilizing locking-jaw mechanism to prevent slippage during tensile loading. Install a four (4) inch extensometer to a conveniently exposed area on the sample. Apply a tensile load uniformly until the 3000 lb. load is attained. Maintain the load for one (1) minute. Measure and record the elongation. Remove the tensile load and measure and record the relaxed elongation. Failure of the tensile load test shall be cause for rejection (see 3.5.5).

4.5.8.6 Bonded seam - in-line tensile load. A sample of the Bladder bonded seam shall be subjected to a tensile load test to verify that the workmanship and bonding techniques are compliant with the requirements of this specification. Failure of the tensile load test shall be cause for rejection (see 3.5.6). Cut a 40-inch sample of seam from the Bladder including enough material to encompass all bonded material at the seam. Install in a tensile test configuration to prevent slippage during tensile loading. Apply a tensile load uniformly until the 3000-pound load is attained. Maintain the load for one (1) minute. Measure and record any separation of the seam area.

4.5.8.7 Bonded seam - across seam tensile load. A sample of the Bladder bonded seam shall be subjected to a tensile load test to verify that the workmanship and bonding techniques are

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compliant with the requirements of this specification. Cut a 12-inch sample of seam from the destructive test Bladder including enough material on both sides of the seam to allow clamping. Install in a tensile test configuration to enable testing across the seam. Apply a tensile load uniformly until the 3000-pound tensile load is attained. Maintain the load for one (1) minute. Measure and record any separation of the seam area. Failure of the tensile load test shall be cause for rejection (see 3.5.7).

4.5.8.8 Bonded fittings (vent-fill/evacuate flanges). A sample of the Bladder bonded fitting shall be subjected to a tensile load test to verify that the workmanship and bonding techniques are compliant with the requirements of this specification. The vent-fill/evacuate flanges shall be removed from the Bladder allowing a minimum of six (6) inches of Bladder material beyond any bonded attachment. Install the flange in a tensile test configuration, such that the Bladder material on opposite sides of the flange are subjected to the tensile load, thereby transmitting the load through the bonded areas of the Bladder and to the flange/Bladder interface. Apply the tensile load uniformly until the 350-pound tensile load is attained. Maintain the load for duration of one (1) minute. Remove the load and perform a visual examination. Failure of the tensile load test shall be cause for rejection (see 3.5.8).

4.5.8.9 "D" rings - tensile load. The Bladder "D" rings shall be subjected to a tensile load test to verify that their strength is compliant with the requirements of this specification. Remove one of each type "D" ring from the Bladder (Lateral-Double "D" Ring and Forward Restraint "D" ring). Install each ring individually in a tensile test machine. Apply a tensile load uniformly to each ring from base to nose until the 10,000-pound tensile load is attained. Maintain the load for 30 seconds. Remove the tensile load and perform a visual examination. Failure of the tensile load test shall be cause for rejection (see 3.5.9).

4.5.8.10 "D" rings - metallurgy. The Bladder "D" ring shall be subjected to a materials analysis to ensure that the material is compliant with the requirements of this specification. Remove one (1) of each type "D" ring from the Bladder restraint harness, (forward restraint "D" ring, and lateral tie-down double "D" ring). Cut a one (1) inch sample from each ring. Place each sample individually into an arc emission spectrometer interconnected with computer interface and printer. Perform an arc emission spectrometry test on each sample and retrieve a data printout. Failure of the analysis shall be cause for rejection (see 3.5.10).

4.5.8.11 Nozzle assemblies - metallurgy. The Bladder nozzle assemblies shall be subjected to a materials analysis to ensure that the materials and protective coatings are compliant with the requirements of this specification. Remove from one (1) fill/evacuate nozzle and one (1) vent nozzle, approximately one (1) square inch of material to be analyzed. Remove paint or any protective coating to present a clean sample. Perform a hardness test on each sample. Following the hardness test, place each sample individually into an arc emission spectrometer, interconnected with the computer interface and printer. Perform an arc emission spectrometry test on each sample and retrieve a data printout. Failure of the analysis shall be cause for rejection (see 3.5.11).

4.5.8.12 Vent and fill/evacuate flanges - metallurgy. The Bladder vent and fill/evacuate flanges shall be subjected to a materials analysis to ensure that the materials and protective coating are compliant with the requirements of this specification. Remove from one (1) vent flange and one

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(1) fill evacuate flange approximately one 1 square inch of material to be analyzed. Remove paint, protective coating, or any bonding material to present a clean sample. Failure of the analysis shall be cause for rejection (see 3.5.12).

4.5.8.13 Tow Ring tensile load. A sample of the Bladder bonded seam shall be subjected to a tensile load test to verify that the workmanship and bonding techniques are compliant with the requirements of this specification. Cut a section of Bladder materiel which encompasses the area of the bonded strap and clamp in a fixture. Install in a tensile test configuration to enable a load of 1500 pound force to be applied. Maintain the load for 1 minute. Measure and record any separation of the nylon strap. Failure of the tensile load test shall be cause for rejection.

5. PACKAGING

5.1Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Bladders covered by this specification are intended for use with the C-130, C-5, and C-17 aircraft.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article is required.

c. Any additional testing requirements.

d. Packaging requirements.

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6.3 Acronyms/abbreviations. This is a listing of acronyms/abbreviations used in this specification.

ABFDS	Aerial Bulk Fuel Delivery System
GPM	Gallons Per Minute
NBR	Acrylonitrile-Butadiene Rubber
NIST	National Institute of Science and Technology
PSIG	Pounds Per Square Inch Gauge
RH	Relative Humidity

6.4 Definitions.

6.4.1 Blister. A blister is a flaw on the surface caused by non-adherence or by separation of an applied substance.

6.4.2 Burr. A burr is a rough edge left on a piece of metal by a cutting or punching tool.

6.4.3 Checking. Checking in coating fabrics will be defined as fine cracking's appearing in the coating compound.

6.4.4 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches. Society of Automotive Engineers (SAE) tool sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

6.4.5 G. A "G" is a unit of acceleration equal to the acceleration due to gravity. It is expressed as a multiples of the units' weight.

6.4.6 Holiday. A holiday in coated fabrics will be defined as a place not covered by coating compound.

6.4.7 Mils. A mil is a unit of length equal to 1/1000 inch.

6.4.8 Pinhole. A pinhole will be defined as minute circular void or solvent blowhole.

6.4.9 Rework. Rework will be defined as an operation performed during a production operation due to an inadequacy or error during that operation.

6.4.10 Repair. Repair will be defined as a corrective operation, which is required because of an omission of a step or incorrect performance of a step or process during a previously completed manufacturing operation.

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6.4.11 Special tools. A tool which is not commercially readily available from a source other than the Bladder contractor.

6.4.12 Wicking. Wicking in coated fabrics will be defined as the absorption of fuel by capillary at attraction.

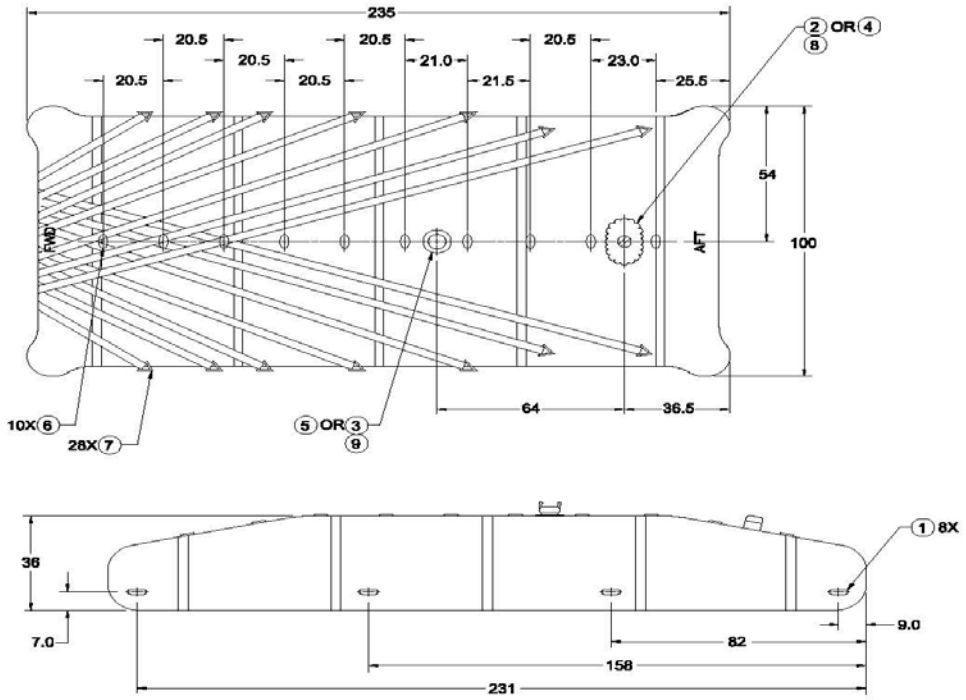
6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, *Shelf-life Management Manual*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/.

6.6 Subject Term (Key Word) Listing.

ABFDS
C-5 Aircraft
C-17 Aircraft
C-130 Aircraft
Cadmium
Hydrocarbon

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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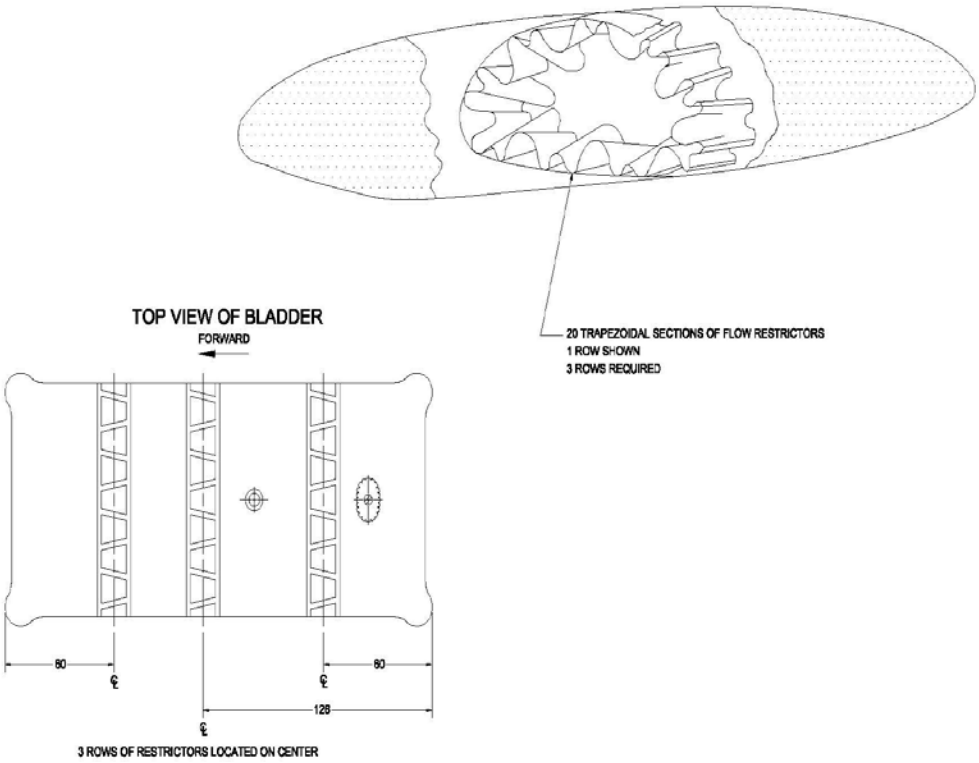
1. ALL DIMENSIONS ARE IN INCHES
   TOLERANCES: DECIMALS
   XXX. - .4
   XX. - .2
   X.X - .5
2. BLADDER SHOWN FILLED TO NORMAL CAPACITY.
3. APPROXIMATE DIMENSIONS OF BLADDER IN FILLED CONDITION. 235 X 100 X 36
4. SHAPE OF CORNERS OPTIONAL
5. UNFILLED BLADDER DIMENSIONS 240 X 108 X 4

FIND	PARAGRAPH	DESCRIPTION	QUANTITY
9	3.3.2.8	Vent Nozzle	1
8	3.3.2.7	Fill / Evacuate Nozzle	1
7	3.3.2.6	D Ring	28
6	3.3.2.6	Double D Ring	10
5	3.3.2.5	Vent Flange Closure Plate	1
4	3.3.2.5	Fill / Evacuate Flange Closure Plate	1
3	3.3.2.4	Vent Flange	1
2	3.3.2.4	Fill / Evacuate Flange	1
1	3.3.2.3	Handle	8
	3.3.1	Bladder Assembly	

Figure 1-A Bladder assembly

FIGURE 1-A. Bladder assembly

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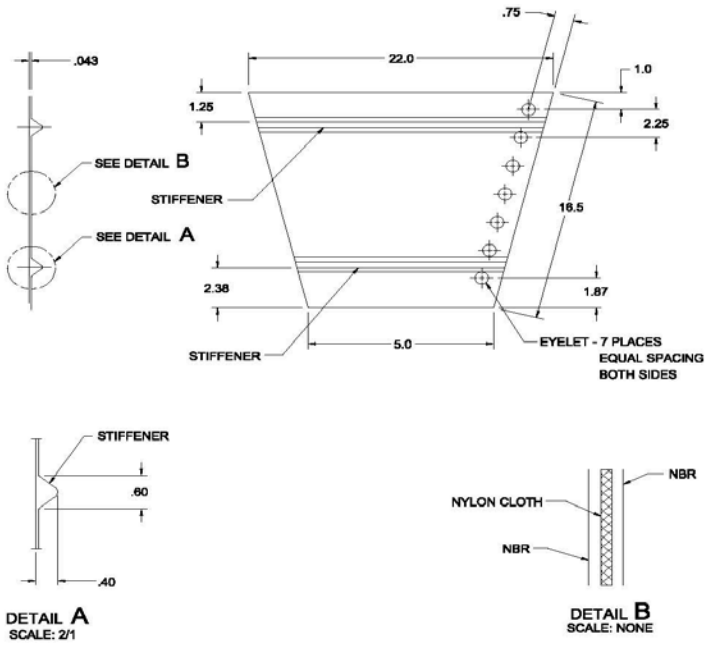


FLOW RESTRICTOR ILLUSTRATION

Figure 2-A

FIGURE 2-A. Flow restrictor illustration

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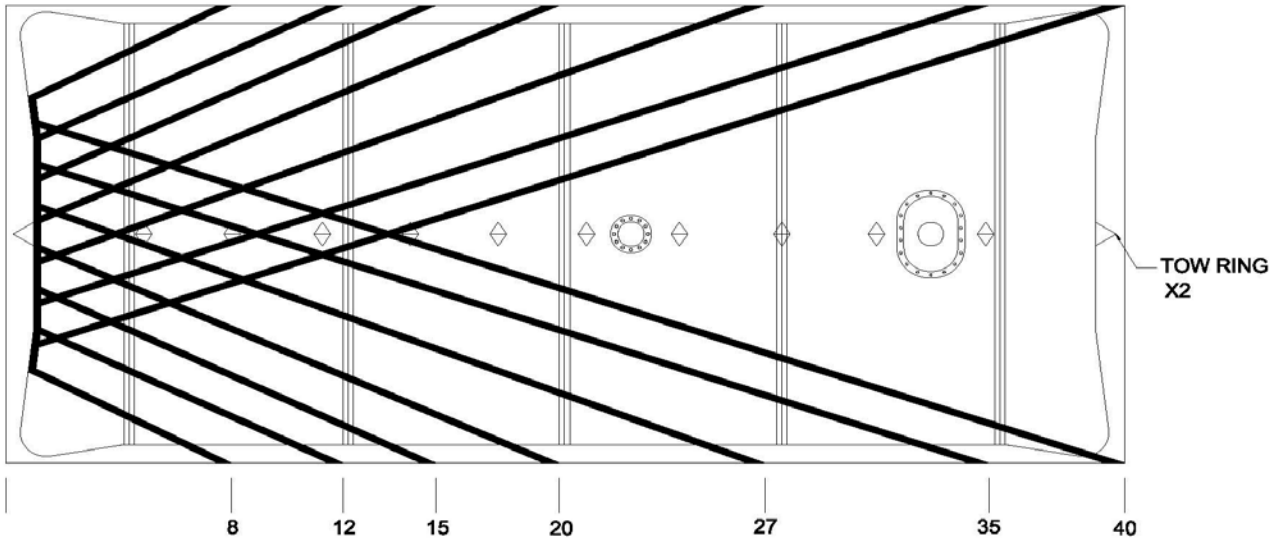
FLUID SURGE SUPPRESOR

MATERIAL: BUNA - ACRYLONITRILE RUBBER AND NYLON CLOTH SANDWICH
DUROMETER READING - 70 SHORE "A"

Figure 3-A

FIGURE 3-A. Fluid surge suppressor

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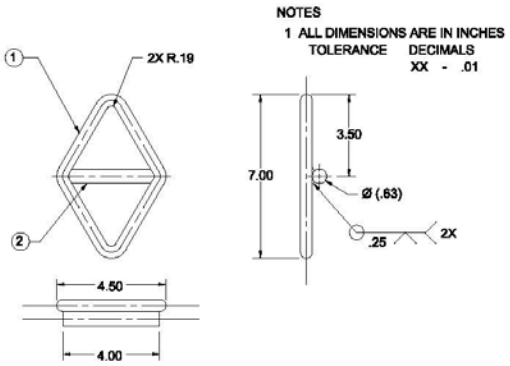


FORWARD HARNESS

Figure 4-A

FIGURE 4-A. Forward harness

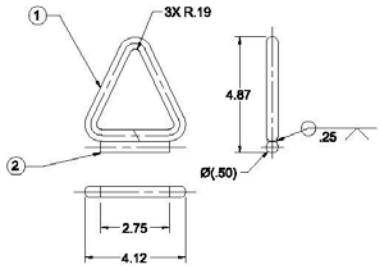
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QTY REQD	FSCM	PART OR IDENTIFING NO.	NOMENCLATURE OR DESCRIPTION	MATERIAL SPECIFICATION	ITEM NO.
1		-03	ROD, ROUND	Ø.63 STAINLESS STEEL ASTM A276-304L	2
1		-01	ROD, ROUND	Ø.50 STAINLESS STEEL ASTM A276-304L	1
		-10	DOUBLE "D" RING		

LIST OF MATERIALS

Figure 5-A. Double "D" ring



QTY REQD	FSCM	PART OR IDENTIFING NO.	NOMENCLATURE OR DESCRIPTION	MATERIAL SPECIFICATION	ITEM NO.
1		-03	ROD, ROUND	Ø.50 STAINLESS STEEL ASTM A276-304L	2
1		-01	ROD, ROUND	Ø.50 STAINLESS STEEL ASTM A276-304L	1
		-10	"D" RING		

LIST OF MATERIALS

Figure 6-A. "D" ring (forward restraint harness)

FIGURE 5-A. Double “D” ring
FIGURE 6-A. “D” ring (forward restraint harness)

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MILITARY INTEREST:

Custodians:
Air Force - 184

Preparing Activity:
Air Force - 184

Reviewers:
Air Force - 99
DLA – CQ

Agent:
Air Force - 99
(Project 5430-2015-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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